

XRES: A Full-Process XR Simulation-Based Educational System for Media Production

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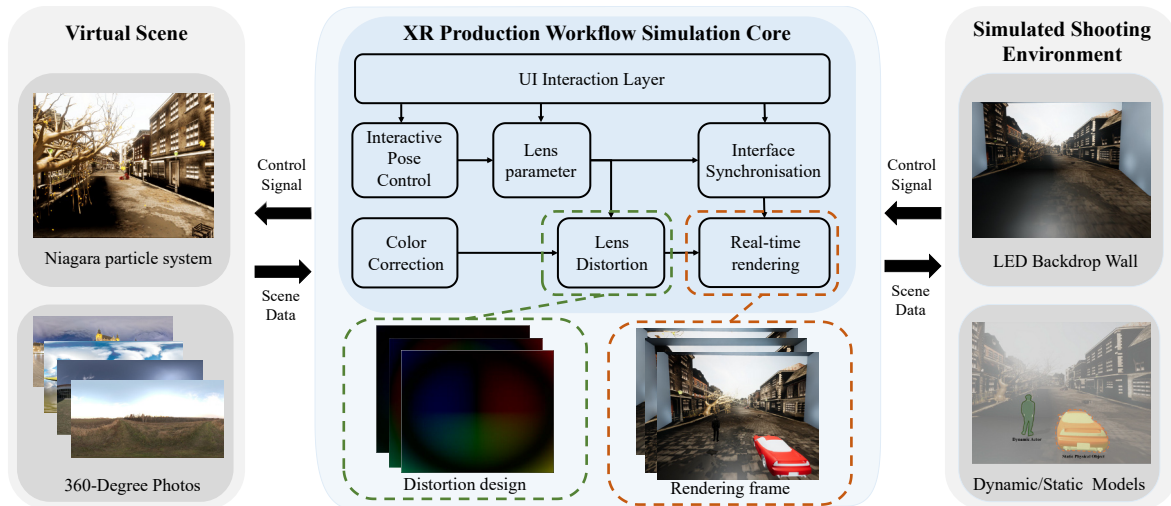


Figure 1: Overview of the XR simulation-based educational system, consisting of a virtual scene, a simulation system, and a simulated shooting environment with synchronized rendering and interaction.

ABSTRACT

Extended Reality (XR) enables immersive media production through virtualized scene creation and real-time rendering, yet professional XR media production workflows remain difficult for novices to access and practice. We present a full-process XR simulation-based educational system that reconstructs the core XR production pipeline within an accessible virtual environment, supporting hands-on learning of camera tracking, real-time rendering, and workflow synchronization. A pilot study comparing PC and VR configurations demonstrates comparable learning effectiveness, with complementary strengths in usability and immersion. These results provide initial evidence for the feasibility of simulation-based XR training in media production education.

Index Terms: Extended Reality, simulation-based education, XR media production.

1 INTRODUCTION

Extended Reality (XR) has transformed media production by enabling immersive storytelling through virtualized scene creation and real-time rendering, reducing reliance on physical locations and sets[2]. While these workflows improve production efficiency, professional XR media production remains technically demanding,

requiring precise camera tracking, lens calibration, real-time rendering control, and color synchronization between physical cameras and LED displays[3]. Due to the high cost and complexity of professional XR facilities, novices have limited opportunities to practice these technical procedures, posing a barrier to skill development.

To address this gap, we propose a full-process XR simulation-based educational system that reconstructs key stages of a professional XR production pipeline within an accessible virtual environment. The system supports hands-on training in virtual scene calibration, camera tracking debugging, real-time rendering adjustment, and multi-interface synchronization, enabling learners to acquire practical skills transferable to real-world XR media production.

2 SYSTEM COMPOSITION

The system follows a three-stage workflow: virtual scene → XR production workflow simulation core → simulated shooting environment, as depicted in Figure 1.

2.1 Virtual Scene

To ensure high-fidelity visual output, the system integrates the Niagara particle system in Unreal Engine 5 (UE5) for dynamic effects. Furthermore, AI-generated 360-degree panoramic images[1] are optimized as peripheral backdrops, balancing real-time rendering performance with environmental realism.

2.2 XR Production Workflow Simulation Core

Based on the full-process operational requirements of XR media production, the system includes three core modules: Camera Tracking, Real-time Rendering, and Real-time Interaction, designed to support experiential learning and hands-on practice of core XR production workflows.

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2.2.1 Camera Tracking Module

Utilizing UE5's Input Mapping Contexts and synchronous transmission, this module coordinates virtual camera movements with simulated physical equipment. By actively manipulating camera poses, learners intuitively grasp the spatiotemporal coupling mechanisms between virtual and physical cameras essential for real-world production.

2.2.2 Real-time Rendering Module

This module processes captured images into texture data to replicate optical and colorimetric characteristics of physical production. LUT-based color correction and precomputed UV offsets are applied to ensure color consistency and simulate lens distortion. Immediate visual feedback allows learners to observe the effects of camera and rendering adjustments, supporting problem-solving skills relevant to real-world XR shooting scenarios.

2.2.3 Real-time Interaction Module

A unified user interface aggregates all system functionalities. This module guides learners through the complex technical pipeline, facilitating the mastery of the systematic full-process workflow.

2.3 Simulated Shooting Environment

The final integrated output is presented in a simulated XR shooting environment consisting of two perpendicular LED screens. Dynamic pedestrian models and static vehicles are added to enhance the clarity and realism of the final visual effect.

3 PILOT STUDY: PRELIMINARY EVALUATION

A pilot study was conducted to gather initial evidence regarding the system's interactivity, immersion, and instructional potential under different device configurations.

3.1 Participants and Apparatus

Ten participants (6 males, 4 females; aged 21–24, $M=22$, $SD=1.05$) from a local university participated. The apparatus included a Windows 11 workstation (24-inch monitor) and a Meta Quest 2 headset.

3.2 Experimental Design and Procedure

Participants were randomly assigned to two groups: **VR Group** (Quest 2) and **PC Group** (Desktop). The procedure consisted of three phases:

- 1. Pre-training Assessment:** To establish a baseline, participants completed a pre-test consisting of 10 multiple-choice questions. These questions assessed their initial proficiency across multiple dimensions of XR production knowledge, including tracking principles and color science.
- 2. Task Execution:** Following a brief tutorial, participants performed a guided simulation task structured around the three core system modules. They were required to: (1) align virtual camera poses (Camera Tracking), (2) correct lens distortion and color consistency (Rendering), and (3) execute the recording workflow to capture a final synchronized clip (Interaction).
- 3. Post-training Evaluation:** Participants completed an identical post-test to measure knowledge gain. Subsequently, they completed subjective questionnaires: System Usability (SU) was measured using a scale adapted from the System Usability Scale (SUS), and Immersion & Presence (IP) was assessed using items adapted from the Igroup Presence Questionnaire (IPQ).

3.3 Results and Discussion

As shown in Figure 2, significant differences were observed in IP ($p < 0.005$) and SU ($p < 0.05$). VR yielded higher immersion scores, benefiting from superior spatial presence and realistic dynamic effects, though constrained by higher system latency and limited usage duration. Conversely, the PC condition demonstrated superior usability, attributed to lower latency and intuitive inputs that facilitate stable parameter control. No significant differences in Learning Effectiveness (LE) and Overall Usage Intention (OUI) ($p > 0.05$) suggest both platforms are instructionally viable.

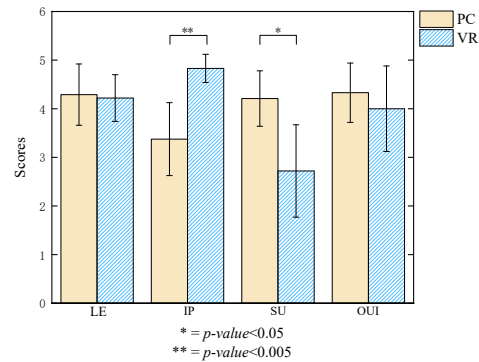


Figure 2: Subjective evaluation scores across PC and VR platforms.

Objective data further substantiated this effectiveness, with test accuracy increasing from 30% to 80% post-training. Overall, results highlight a trade-off: PC favors precise, low-latency interaction, while VR excels in experiential simulation. Despite the limited sample size ($N = 10$), these findings provide initial empirical validation to guide future large-scale experiments.

4 CONCLUSION

This study presents a full-process XR simulation-based educational system for training contemporary media production workflows. By reconstructing the XR production pipeline within an accessible virtual environment, the system enables structured learning without reliance on physical facilities. A pilot study comparing PC and VR platforms demonstrates learning effectiveness and reveals complementary strengths in usability and immersion. These findings suggest that simulation-based XR training can lower learning barriers and inform the design of future XR education systems. Future work will focus on larger-scale evaluations and further refinement of interaction and pedagogical design.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (No.62572047)

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